

Student Number: _____

KAMBALA

MATHEMATICS

2 UNIT

TRIAL HSC EXAMINATION

AUGUST 2007

Time Allowed: 3 hours
Reading Time: 5 minutes

INSTRUCTIONS

- This examination contains 10 questions of equal value. Marks for each question are shown.
- Answer all questions in the writing booklets provided. Start each question in a new booklet.
- Calculators may be used.
- Show all necessary working.
- Marks may not be awarded for careless or badly arranged work.

Question 1 (12 marks) Use a SEPARATE writing booklet

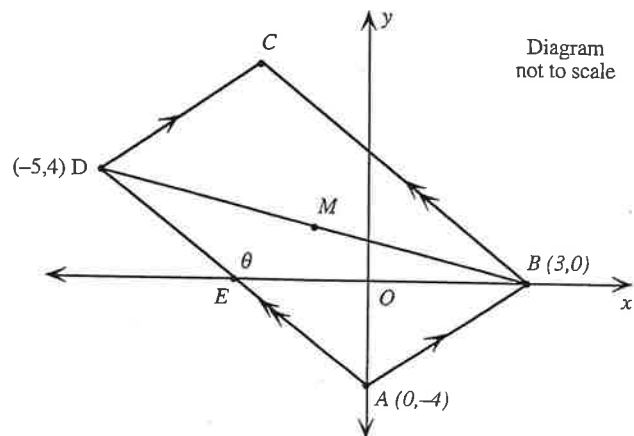
Marks

- (a) Evaluate $\frac{56.07 - \sqrt{39.2}}{7.1}$, correct to 3 significant figures. 2
- (b) Simplify: $\frac{3x-4}{2} - \frac{5-x}{3}$ 2
- (c) If \$5000 is invested for 10 years and interest is compounded at 9% per annum, what is the final value of the investment? 2
- (d) Find the solution of θ , in radians, in the interval $\frac{3\pi}{2} \leq \theta \leq 2\pi$ if $\cos \theta = \frac{\sqrt{3}}{2}$. 2
- (e) Find $\int \sec^2 5x \, dx$. 2
- (f) Find integers a and b such that $\frac{4}{2-\sqrt{3}} = a + b\sqrt{3}$. 2

Question 2 (12 marks) Use a SEPARATE writing booklet

Marks

In the number plane diagram below, $A(0,-4)$, $B(3,0)$ and $D(-5,4)$ are vertices of parallelogram $ABCD$. The side AD meets the x -axis at E . $\angle DEB = \theta$ and M is the midpoint of the diagonal BD .



- Find the gradient of the line AD .
- Find θ to the nearest degree.
- Show that AB has equation $4x - 3y - 12 = 0$.
- Find the perpendicular distance between D and the line AB .
- Find the area of the parallelogram $ABCD$.
- Find the co-ordinates of M , the midpoint of BD .
- Hence, or otherwise, find the co-ordinates of C .

1
1
2
2
2
2
2

Question 3 (12 marks) Use a SEPARATE writing booklet

Marks

(a) Differentiate: $y = x \sin 5x$

2

(b) Differentiate: $y = \frac{e^{5x+4}}{2x^2}$

2

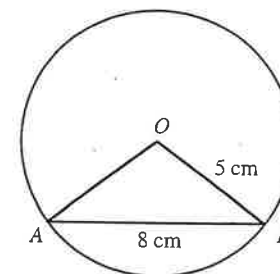
(c) (i) Given that $y = \sqrt{4x+9}$, show that $\frac{dy}{dx} = \frac{2}{3}$ when $x = 0$.

2

(ii) Hence determine the equation of the normal to the curve $y = \sqrt{4x+9}$ at the point $(0,3)$ on it.

2

(d)



AB is a chord of length 8 cm in a circle centre O of radius 5 cm.

(i) Using the cosine rule, find the size of $\angle AOB$.

2

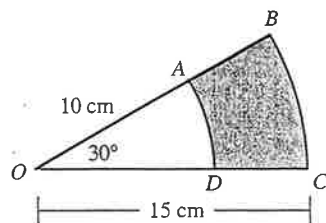
(ii) Find the length of the minor arc AB . Give your answer correct to 2 decimal places.

2

Question 4 (12 marks) Use a SEPARATE writing booklet

Marks

- (a) Find $\int x - 5 \sin 3x \, dx$. 2
- (b) Evaluate the definite integral $\int_0^1 (e^{2x} + 1) \, dx$, leaving the answer in exact form. 2
- (c) Consider the equation $2x^2 - (k + 3)x + 2 = 0$. For what values of k does the equation have real roots? 3
- (d) 2



In the diagram above, AD and BC are arcs of concentric circles with O as centre. Calculate, in terms of π , the area of the shaded region $ABCD$.

- (e) Consider the parabola $(x - 1)^2 = 12y$.
- (i) Find the co-ordinates of its vertex and focus. 2
- (ii) Illustrate this information in a neat sketch. 1

Question 5 (12 marks) Use a SEPARATE writing booklet

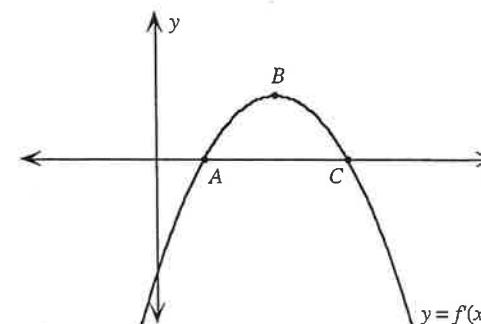
Marks

- (a) (i) By completing the square, sketch the graph of $x^2 + 6x + y^2 = 0$. 2
- (ii) State the range of the graph in part (i). 1
- (b) For the function $y = x^3 - 3x^2 - 9x + 6$:
- (i) Find the co-ordinates of any stationary points and determine their nature. 3
- (ii) Find the co-ordinates of any points of inflection. 2
- (iii) At what point does the curve cut the y -axis? 1
- (iv) For what values of x is the curve concave up? 1
- (v) Sketch the curve, showing all essential features. 2

Question 6 (12 marks) Use a SEPARATE writing booklet

Marks

- (a) Evaluate: $\int_0^{\ln 2} e^{-x} \, dx$ 2
- (b) Evaluate: $\sum_{n=3}^6 (n + 1)^2$ 1
- (c) Solve: $e^{2x} + 5e^x - 6 = 0$ 3
- (d) 2



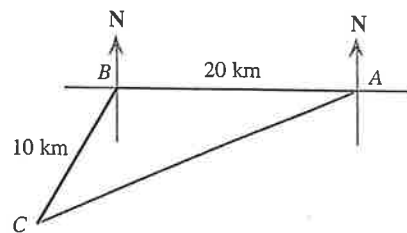
The diagram above shows the graph of $y = f'(x)$, where $f'(x)$ is the derivative of $f(x)$. What point (A , B or C) on this graph corresponds to a maximum turning point of the function $y = f(x)$? Give reasons for your answer.

- (e) Josie accepted a job that pays an initial salary of \$35 000 per annum. After each year of service she will receive an increment of \$5000 until she reaches a maximum salary of \$105 000.
- (i) What will be her salary after 5 years of service? 2
- (ii) How long will she have to work before she has earned a total of \$540 000? 2

Question 7 (12 marks) Use a SEPARATE writing booklet

Marks

(a)



In the diagram above, a hiker walks 20 km due west from A to B and then another 10 km in a direction 220°T to C.

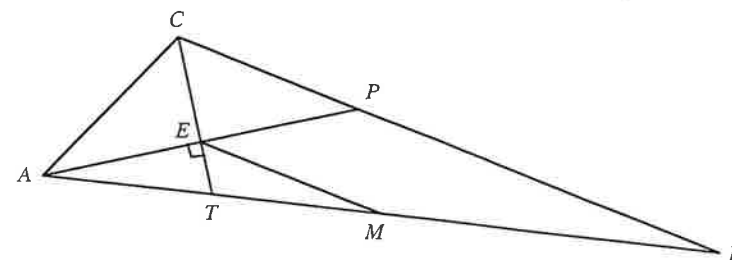
- (i) Explain why $\angle ABC = 130^\circ$. 1
 - (ii) Find the distance and bearing of the hiker from A. 3
- (b) At the beginning of summer, a dam is 56% full. Due to evaporation and use by consumers it loses 8% of the water in the dam each week. There is no further inflow of water during the season.
- (i) What percentage of the full dam remains after 4 weeks? 2
 - (ii) If water rationing will be introduced when the dam is 30% full, find when rationing commences. 2
- (c) Consider the function $y = \ln(x + 3)$ for $x > -3$.
- (i) Sketch the function, showing its essential features. 2
 - (ii) Use Simpson's Rule with three function values to find an approximation to $\int_{-1}^1 \ln(x + 3) dx$. 2

Question 8 (12 marks) Use a SEPARATE writing booklet

Marks

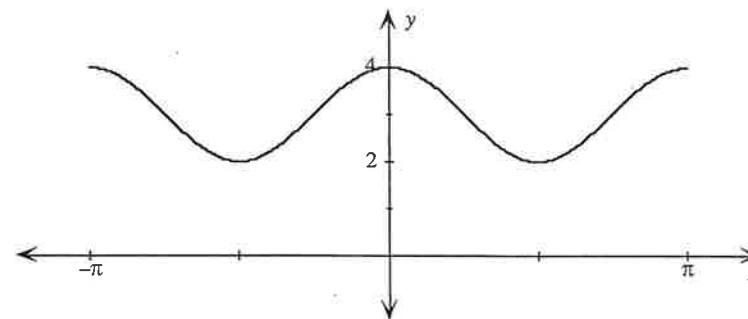
- (a) Calculate the exact volume generated when the region enclosed by the curve $y = 2 + 3e^{-x}$ for $0 \leq x \leq 2$ is rotated about the x -axis. 3
- (b) A tap that empties the water from an urn is turned on slowly so that the volume flow rate, $\frac{dV}{dt}$, varies with time according to the relation $\frac{dV}{dt} = qt^2$, where $t > 0$ and q is a constant. Calculate the total volume of water that flows through the tap in the first 5 seconds if $q = 3.6 \text{ m}^3\text{s}^{-2}$. 2

(c)



In the diagram above, CT bisects $\angle ACB$. M is the midpoint of AB and AE is perpendicular to CT . AE produced meets BC at P .

- (i) Copy the diagram into your answer book and mark in all the given information. 2
 - (ii) Prove that $\triangle ACE$ is congruent to $\triangle PCE$. 1
 - (iii) Explain why $AE = EP$. 1
 - (iv) State why $\triangle AEM$ is similar to $\triangle APB$ and hence prove that EM is parallel to PB . 2
- (d) 2



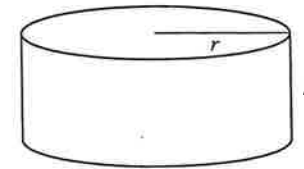
The diagram above shows the graph of $y = \cos 2x + 3$ from $-\pi \leq x \leq \pi$. Find the area under the curve $y = \cos 2x + 3$ between $x = 0$ and $x = \frac{7\pi}{12}$. Express your answer in terms of π .

Question 9 (12 marks) Use a SEPARATE writing booklet**Marks**

- (a) The number of bacteria (N) in a culture is growing exponentially according to the formula $N = 80e^{kt}$.
- (i) What is the initial number of bacteria? 1
 - (ii) After eight hours the number of bacteria has doubled. Calculate the value of k . Give your answer correct to 3 decimal places. 2
 - (iii) How many bacteria, correct to the nearest ten, will there be after 12 hours? 1
 - (iv) At what rate will the bacteria be increasing after 12 hours? 1
- (b) The position of a particle moving along the x -axis is given by $x = 2t + e^{-2t}$ where t is the time in seconds and x is measured in centimetres.
- (i) Show that the particle is at rest when $t = 0$. 2
 - (ii) What is the acceleration of the particle after 1 second? 2
 - (iii) What is the limiting velocity of the particle? 1
- (c) If $y = \log_e x$:
- (i) express x in terms of y and e . 1
 - (ii) express $\log_{10} x$ in terms of y . 1

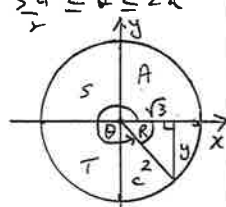
Question 10 (12 marks) Use a SEPARATE writing booklet**Marks**

- (a) A closed cylindrical can of radius r cm and height h cm has a capacity of $16\pi \text{ cm}^3$.



- (i) Show that the height can be expressed as $h = \frac{16}{r^2}$. 1
 - (ii) Show that the surface area of the cylinder is given by $S = 2\pi r^2 + \frac{32\pi}{r}$. 1
 - (iii) What are the radius of the base and the height of the cylinder for the total surface area to be a minimum? 3
- (b) Alicia borrows \$100 000 at a reducible interest rate of 7.2% p.a., compounded monthly. The debt is to be repaid with monthly repayments over 10 years.
- (i) If Alicia pays \$ P per month, show that after 2 months, the amount owing is $\$[100\,000(1.006)^2 - P(1.006) - P]$. 2
 - (ii) The monthly repayment is \$1172. How much will Alicia still owe after 3 years? 2
 - (iii) After 3 years, the bank tells Alicia that the interest rate has increased by 0.6% to 7.8% p.a. How much will her new monthly payment need to be in order to still pay out the loan at the end of the 10 years? 3

END OF EXAMINATION

Qn	Solutions	Marks	Comments+Criteria
1(a)	$\frac{56.07 - \sqrt{39.2}}{7.1} = \frac{49.809...}{7.1}$ $= 7.01535$ $= \boxed{7.02}$	1 1	correct answer rounding
(b)	$\frac{3x-4}{2} - \frac{5-x}{3} = \frac{3(3x-4) - 2(5-x)}{6}$ $= \frac{9x - 12 - 10 + 2x}{6}$ $= \frac{11x - 22}{6}$	1 1	
(c)	$P = \$5000, n = 10 \text{ yrs}, r = 9\% \text{ pa}$ $A = P \times (1+r)^n$ $= 5000 \times (1.09)^{10}$ $\boxed{FV = \$11,836.82}$	1 1	
(d)	$\cos \theta = \frac{\sqrt{3}}{2}, \quad \frac{3\pi}{2} \leq \theta \leq 2\pi$ $\cos 30^\circ = \sqrt{3}/2$ $\therefore \theta = 2\pi - \pi/6$ $\boxed{\theta = \frac{11\pi}{6}}$ 	1 1 1	1 for link to acute angle 1 for correct angle in radians
(e)	$\int \sec^2 x \, dx = \frac{1}{5} \tan 5x + C$	1 1	1 for tan 1 for correct coefficients
(f)	$\frac{4}{2-\sqrt{3}} = a + b\sqrt{3}$ $\frac{4}{2-\sqrt{3}} \times \frac{2+\sqrt{3}}{2+\sqrt{3}} = a + b\sqrt{3}$ $\frac{8+4\sqrt{3}}{1} = a + b\sqrt{3}$ $\therefore \boxed{a=8, b=4}$	1 1 1	1 for rationalising 1 for equating coefficients

Qn	Solutions	Marks	Comments+Criteria
2(i)	$A = (0, -4), B = (-5, 4)$ $x_1, y_1 \quad x_2, y_2$ $m = \frac{y_2 - y_1}{x_2 - x_1}$ $= \frac{4 - (-4)}{-5 - 0}$ $= \frac{8}{-5}$ $\boxed{m = -8/5}$	1	correct answer
(ii)	$m = \tan \theta$ $\tan \theta = -8/5$ $\therefore \theta = 180^\circ - 58^\circ$ $\boxed{\theta = 122^\circ}$	1	
(iii)	$A(0, -4), B(3, 0)$ $x_1, y_1 \quad x_2, y_2$ $\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$ $\frac{y + 4}{x - 0} = \frac{0 + 4}{3 - 0}$ $3y + 12 = 4x$ $\therefore AB: \boxed{4x - 3y - 12 = 0}$	1	1 for gradient 1 for substitution of point into formula
(iv)	$D = (-5, 4)$ $d_{D \rightarrow AB} = \frac{ 4(-5) - 3(4) - 12 }{\sqrt{(4)^2 + (-3)^2}}$ $\boxed{d_{D \rightarrow AB} = \frac{44}{5} \text{ units}}$	1	substitution into formula correct answer
(v)	$d_{AB} = \sqrt{(3-0)^2 + (0+4)^2}$ $d_{AB} = 5$ $\therefore \text{Area}(ABCD) = b \times h$ $= 5 \times 44/5$ $\boxed{\text{Area} = 44 \text{ u}^2}$	1	working such as d_{AB} correct answer

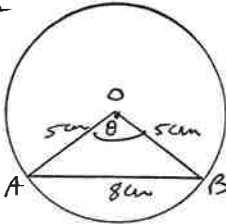
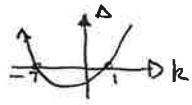
(3)

Qn	Solutions	Marks	Comments+Criteria
2(vi)	$B = (3, 0), D = (-5, 4)$ $M = \left(\frac{3 + (-5)}{2}, \frac{0 + 4}{2}\right)$ $M = (-1, 2)$	1 1	substitution correct answer
(vii)	$M(AC) = M(BD)$ Let $C = (x, y)$ $\left(\frac{x+0}{2}, \frac{y-4}{2}\right) = (-1, 2)$ $\therefore \frac{x+0}{2} = -1$ $x = -2$ $\frac{y-4}{2} = 2$ $y-4 = 4$ $y = 8$ $\therefore C = (-2, 8)$		1 for working 1 for answer

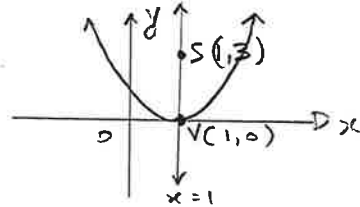
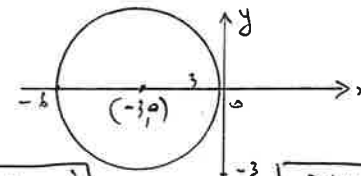
(4)

Qn	Solutions	Marks	Comments+Criteria
3(a)	$y = x \cdot \sin 5x$ $y' = \sin 5x \cdot 1 + x \cdot 5 \cos 5x$ $y' = \sin 5x + 5x \cos 5x$	2	1 for use of product rule 1 for correct substitution
(b)	$y = \frac{e^{5x+4}}{2x^2}$ $y' = \frac{(2x) \cdot 5e^{5x+4} - e^{5x+4}(4x)}{4x^4}$ $y' = \frac{e^{5x+4}(5x-2)}{2x^3}$	2	1 for use of quotient rule 1 for correct substitution
(c)(i)	$y = \sqrt{4x+9} = (4x+9)^{1/2}$ $y' = \frac{1}{2}(4x+9)^{-1/2} \cdot 4$ $y' = \frac{2}{\sqrt{4x+9}}$ when $x=0$, $y' = \frac{2}{\sqrt{9}} = \frac{2}{3}$	1 1	
(ii)	Normal is $y-4 = m(x-1)$ At $(0, 3)$ $m = -\frac{3}{1}$ $y-3 = -\frac{3}{1}(x-0)$ $y-6 = -3x$ $3x + 2y - 6 = 0$	1	1 for gradient of normal 1 for substitution into equation of line

5

Qn	Solutions	Marks	Comments+Criteria
3(d)(i)	$\cos \theta = \frac{b^2 + c^2 - a^2}{2bc}$ $\cos \theta = \frac{5^2 + 5^2 - 8^2}{2 \times 5 \times 5}$ $= \frac{-14}{50}$ $\therefore \theta = 106^\circ 16'$ 	1 1	substitution into formula angle (rounding ignored)
(ii)	$\theta = 106^\circ 16' = 1.85459 \dots$ $l = r \times \theta$ $l = 5 \times 1.85459 \dots$ $l = 9.27295 \dots$ $l = 9.27 \text{ cm}$		1 for conversion to radians 1 for substitution into formula (ignore rounding)
4(a)	$\int x - 5 \sin 3x \, dx = \frac{x^2}{2} + \frac{5}{3} \cos 3x + C$	2	1 for $\frac{x^2}{2}$ 1 for $\frac{5}{3} \cos 3x$ ignore + C integration
(b)	$\int_0^1 (e^{2x} + 1) \, dx = \left[\frac{1}{2} e^{2x} + x \right]_0^1$ $= \left(\frac{1}{2} e^2 + 1 \right) - \left(\frac{1}{2} + 0 \right)$ $= \frac{1}{2} (e^2 + 1)$ or $\frac{1}{2} e^2 + \frac{1}{2}$	1 1	Must be in exact form
(c)	$2x^2 - (k+3)x + 2 = 0$ For real roots $\Delta \geq 0$ $\Delta = b^2 - 4ac$ $= [-(k+3)]^2 - 4(2)(2)$ $= k^2 + 6k + 9 - 16$ $\Delta = k^2 + 6k - 7$ $\Delta \geq 0$ $(k+7)(k-1) \geq 0$ $\therefore k \leq -7, k \geq 1$ 	1 1	

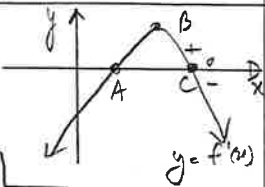
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Qn	Solutions	Marks	Comments+Criteria
4(d)	$A_s = \frac{1}{2} R^2 \theta - \frac{1}{2} r^2 \theta$ $= \frac{1}{2} (15)^2 \frac{\pi}{6} - \frac{1}{2} (10)^2 \frac{\pi}{6}$ $= \frac{225\pi}{12} - \frac{100\pi}{12}$ $A_s = \frac{125\pi}{12} \text{ cm}^2$	1 1	
(e)	$(x-1)^2 = 12y$ $(x-1)^2 = 4(3)(y-0)$		
(i)	$V(1, 0)$ $a = 3$ $S = (1, 3)$	1 1	
(ii)		1	correct graph showing answers above
5(a)(i)	$x^2 + 6x + y^2 = 0$ $x^2 + 6x + 9 + y^2 = 9$ $\therefore (x+3)^2 + (y-0)^2 = 9$ This is a circle, centre $(-3, 0)$ and radius 3 units	1	
		1	graph from above answer
(ii)	Range is $-3 \leq y \leq 3$	1	

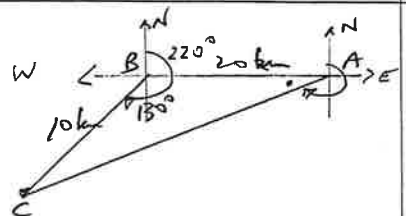
Qn	Solutions	Marks	Comments+Criteria									
5(b)	$y = x^3 - 3x^2 - 9x + 6$ (i) $y' = 3x^2 - 6x - 9$ For a stat. pt., $y' = 0$ $0 = 3x^2 - 6x - 9$ $0 = x^2 - 2x - 3$ $0 = (x-3)(x+1)$ $\therefore x = 3, -1$ when $x = 3$, $y = 27 - 27 - 27 + 6 = -21$ when $x = -1$, $y = -1 - 3 + 9 + 6 = 11$ $\therefore$ stat pts are $(3, -21), (-1, 11)$ $y'' = 6x - 6$ when $x = 3$, $y'' = 18 - 6 = 12 > 0$ $\therefore (3, -21)$ is a MIN TURN PT when $x = -1$, $y'' = -6 - 6 = -12 < 0$ $\therefore (-1, 11)$ is a MAX TURN-PT (ii) For an inflection pt, $y'' = 0$ and changes sign. $y'' = 6x - 6$ $6x - 6 = 0$ $\therefore x = 1$ when $x = 1$, $y = 1 - 3 - 9 + 6 = -5$ Test <table border="1"> <tr> <td>x</td> <td>0</td> <td>1</td> <td>2</td> </tr> <tr> <td>y''</td> <td>-6</td> <td>0</td> <td>12</td> </tr> </table> $\therefore$ Inflection pt. at $(1, -5)$	x	0	1	2	y''	-6	0	12			1 for x-values 1 for y-values 1 for testing
x	0	1	2									
y''	-6	0	12									
			1 for co-ordinate 1 for testing									

Qn	Solutions+Criteria	Marks	Comments+Criteria
5(iii)	$y = x^3 - 3x^2 - 9x + 6$ when $x = 0$, $y = 6$. $\boxed{(0, 6)}$	1	
(iv)	For concave up, $y'' > 0$. $y'' = 6x - 6$ $\therefore 6x - 6 > 0$ $\therefore \boxed{x > 1}$	1	
(v)			1 for shape 1 for key points
6(a)	$\int_0^{\ln 2} e^{-x} dx = [-e^{-x}]_0^{\ln 2}$ $u = -[e^{-\ln 2} - e^0]$ $u = -[e^{\ln 1/2} - 1]$ $u = -[\frac{1}{2} - 1]$ $u = \boxed{1/2}$	1	
(b)	$\sum_{n=3}^6 (n+1)^2 = 4^2 + 5^2 + 6^2 + 7^2$ $= \boxed{126}$	1	
(c)	$e^{2x} + 5e^x - 6 = 0$ Let $m = e^x$ $m^2 + 5m - 6 = 0$ $(m+6)(m-1) = 0$ $\therefore m = -6, m = 1$ $\therefore e^x \neq -6, \therefore e^x = 1 \therefore \boxed{x = 0}$	2	

(9)

Qn	Solutions	Marks	Comments+Criteria
6(d)	For a max. turn. pt, $y' = 0$ and $y' = +0 -$ Max. Turning Pt for $y = \tan^{-1} x + C$ 		1 point 1 reason
ce)	AP $\begin{cases} a = \\$35,000 \\ d = \\$5,000 \end{cases}$ (i) $TS = a + 4d$ $u = \\$35,000 + 4 \times \\$5,000$ $TS = \\$55,000$ (ii) $S_n = \\$540,000$ $\frac{n}{2} [2a + (n-1)d] = 540,000$ $\frac{n}{2} [70,000 + (n-1)5000] = 540,000$ $\frac{n}{2} [5000n + 65000] = 540,000$ $n(5000n + 65000) = 1,080,000$ $5000n^2 + 65000n - 1,080,000 = 0$ $5n^2 + 65n - 1080 = 0$ $n^2 + 13n - 216 = 0$ $n = \frac{-13 \pm \sqrt{13^2 - 4 \cdot 1 \cdot (-216)}}{2}$ $= \frac{-13 \pm \sqrt{1033}}{2} \approx -9.57 \text{ or } 22.57$ $n > 0$ so after 10 years	1	1 for recognising AP
		2	2 mks for solving to $n^2 + 13n - 216 = 0$

(10)

Qn	Solutions	Marks	Comments+Criteria
7(a)	 (i) $\angle ABC = 220^\circ - 90^\circ$ $\angle ABC = 130^\circ$ (ii) $AC^2 = 10^2 + 20^2 - 2 \times 10 \times 20 \cos 130^\circ$ $AC = 27.52 \text{ km}$ $\frac{\sin A}{10} = \frac{\sin 130^\circ}{27.52}$ $\sin A = \frac{10 \times \sin 130^\circ}{27.52}$ $\therefore A = 16^\circ 10'$ $\therefore$ bearing of Hiker from A is Bearing = $270^\circ - 16^\circ$ Bearing = 254° T (b) (i) Beginning 56% full let amount remaining after n weeks be A_n $A_1 = 0.56 \times 0.92$ $A_2 = 0.56 \times 0.92 \times 0.92$ $A_3 = 0.56 \times 0.92^3$ $A_4 = 0.56 \times 0.92^4$ $= 0.40118$ $\therefore 40.1\%$	1	
		1	
		1	
		1	

(14)

Qn	Solutions	Marks	Comments+Criteria
8d)	$A = \int_0^{\frac{\pi}{2}} (\cos 2x + 3x) dx$ $= \left[\frac{1}{2} \sin 2x + \frac{3x^2}{2} \right]_0^{\frac{\pi}{2}}$ $= \left(\frac{1}{2} \sin \frac{\pi}{2} + \frac{3\pi^2}{8} \right) - (0 + 0)$ $= \frac{1}{2} \left(\frac{1}{2} \right) + \frac{3\pi^2}{8}$ $\therefore A = \frac{3\pi^2}{8} + \frac{1}{4}$ $\boxed{A = \frac{1}{4} (7\pi^2 - 1) \text{ u}^2}$	1	
9(a)	$N = 80 e^{kt}$ (i) when $t=0$, $N = 80 e^0$ $\boxed{N = 80}$ (ii) when $t=8$, $N = 160$. $= 160 = 80 e^{8k}$ $e^{8k} = 2$ $8k = \ln 2$ $\boxed{k = \frac{\ln 2}{8} = 0.08664}$ $\therefore \boxed{k = 0.087} \text{ (3 d.p.)}$ (iii) when $t=12$, $N = 80 e^{0.087 \times 12}$ $= 226.27$ $\boxed{N = 230} \text{ (tnt)}$ (iv) $dN/dt = k \times N$ $= 0.087 \times 230$ $\boxed{\text{Rate} = 19.66/\text{h}} \text{ (using exact values)}$ or 20.016/h (using answers from (ii) + (iii))	1	ignore rounding.

(15)

Qn	Solutions	Marks	Comments+Criteria
9(b)	$x = 2t + e^{-2t}$ (i) $\frac{dx}{dt} = 2 - 2e^{-2t}$ when $t=0$, $\frac{dx}{dt} = 2 - 2 \times e^0$ $= 2 - 2$ $\boxed{\frac{dx}{dt} = 0}$ $\therefore$ Particle is at rest when $t=0$. (ii) $\frac{d^2x}{dt^2} = +4e^{-2t}$ when $t=1$, $\boxed{\frac{d^2x}{dt^2} = 4e^{-2} \text{ cm/s}^2}$ (iii) $\lim_{t \rightarrow \infty} \frac{dx}{dt} = \lim_{t \rightarrow \infty} \left(2 - \frac{2}{e^{2t}} \right)$ $= 2 - 0$ $\therefore \boxed{\lim \text{vel} = 2 \text{ cm/s}}$ (c) $y = \log_e x$ (i) $\boxed{x = e^y}$ (ii) $\log_{10} x = \log_{10} e^y$ $\therefore \boxed{\log_{10} x = y \log_{10} e}$	1	

(16)

Qn	Solutions	Marks	Comments+Criteria
10(a)(i)	$V = \pi r^2 h$ $V = 16\pi$ $\therefore \pi r^2 h = 16\pi$ $\therefore h = \frac{16}{r^2}$	1	
(ii)	$S = 2\pi r^2 + 2\pi r h$ $u = 2\pi r^2 + 2\pi r \left(\frac{16}{r^2}\right)$ $\therefore S = 2\pi r^2 + \frac{32\pi}{r}$	1	
(iii)	for a Min, $S' = 0$, $S'' > 0$. $S = 2\pi r^2 + 32\pi r^{-1}$ $S' = 4\pi r - \frac{32\pi}{r^2}$ $0 = 4\pi r - \frac{32\pi}{r^2}$ $\therefore 4\pi r = \frac{32\pi}{r^2}$ $r^3 = 8$ $\therefore r = 2 \text{ cm}$ $S' = 4\pi r - 32\pi r^{-2}$ $S'' = 4\pi + 64\pi r^{-3}$ $= 4\pi + \frac{64\pi}{r^3}$ $r=2: S'' = 4\pi + \frac{8\pi}{1} = 12\pi > 0 \therefore \text{Min } S$ $r=2, h = \frac{16}{4} = 4$ $\therefore r=2\text{cm}, h=4\text{cm for Min } S$	1	1 for testing

(17)

Qn	Solutions	Marks	Comments+Criteria
10(b)	$P = \$100,000$ $r = 7.2\% \text{ p.a.} = 0.072 \text{ p.a.}$ $r = 0.006$ $n = 10 \text{ yrs} = 10 \times 12 = 120 \text{ months}$		
(i)	$A_n = \text{amount owing after } n \text{ months.}$ $A_1 = 100,000(1.006) - P$ $A_2 = A_1(1.006) - P$ $= (100,000(1.006) - P)(1.006) - P$ $A_2 = 100,000(1.006)^2 - P(1.006) - P$	1	Derivation of A_2 had to be clear
(ii)	$n = 3 \text{ years} = 36 \text{ months.}$ $A_{36} = (100,000(1.006)^{36} - 1172 \times (1 + 1.006 + 1.006^2 + \dots + 1.006^{35}))$ $A_{36} = 100,000(1.006)^{36} - 1172 \times 536$ $S_{36} = \frac{1.((1.006)^{36} - 1)}{1.006 - 1}$ $S_{36} = 40.050$ $\therefore A_{36} = 100,000(1.006)^{36} - 1172 \times 40.050$ $A_{36} = \$77,091.25$ $\therefore \$77,091.25$	1	

(18)

Qn	Solutions	Marks	Comments + Criteria
10 (b)	(iii) After 3 years $A_{36} = 77\,091.25$ She has 7 more years to go Let amount owing n months after the 3 years be $\\$B_n$ $r = 0.078 \div 12 = 0.0065$ $B_n = 77\,091.25 (1.0065)^n - M(1 + 1.0065 + \dots + 1.0065^{n-1})$ 7 years later $B_{84} = 0$ $77\,091.25 (1.0065)^{84} - M \underbrace{(1 + 1.0065 + \dots + 1.0065^{83})}_S = 0 \quad -1$ $S = \frac{r(r^n - 1)}{r - 1}$ $= \frac{1(1.0065^{84} - 1)}{1.0065 - 1}$ $= 111.27$ $77\,091.25 (1.0065)^{84} \cdot MS = 0$ $M = \frac{77\,091.25 (1.0065)^{84}}{S}$ $= 1193.89$ $\therefore$ Her repayment needs to be \$1193.89		